

JavaScript WeakSet

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Creating a WeakSet

A `WeakSet` is a collection of objects. Unlike a `Set`, a `WeakSet` can only contain **objects** (not primitive values like strings or numbers):

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript WeakSet</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Creating a WeakSet</h4>
<p id="demo"></p>

<script>
// Create a WeakSet with objects
const obj1 = { name: "John" };
const obj2 = { name: "Jane" };
const obj3 = { name: "Bob" };

const weakSet = new WeakSet([obj1, obj2, obj3]);

// Add another object
const obj4 = { name: "Alice" };
weakSet.add(obj4);

document.getElementById("demo").innerHTML =
  "WeakSet created with objects<br>" +
  "has(obj1): " + weakSet.has(obj1) + "<br>" +
  "has(obj4): " + weakSet.has(obj4) + "<br><br>" +
  "Note: WeakSet can only hold objects,<br>" +
  "not strings, numbers, or other primitives";
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

WeakSet Methods

A WeakSet supports only three methods: `add()` , `delete()` , and `has()` :

```

<!DOCTYPE html>
<html>
<body>

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<h4>WeakSet Methods</h4>
<p id="demo"></p>

<script>
const weakSet = new WeakSet();

const obj1 = { id: 1 };
const obj2 = { id: 2 };

// add() - add an object
weakSet.add(obj1);
weakSet.add(obj2);

let text = "After adding 2 objects:<br>";
text += "has(obj1): " + weakSet.has(obj1) + "<br>";
text += "has(obj2): " + weakSet.has(obj2) + "<br><br>";

// delete() - remove an object
weakSet.delete(obj1);
text += "After delete(obj1):<br>";
text += "has(obj1): " + weakSet.has(obj1) + "<br><br>";

text += "<b>Methods available:</b><br>";
text += "add(obj) - add an object<br>";
text += "delete(obj) - remove an object<br>";
text += "has(obj) - check if object exists<br><br>";

text += "WeakSet has NO size property and NO iteration methods";

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>

```

Example 2

Result [View Example](#)

WeakSet vs Set

The main differences between WeakSet and Set:

Feature	Set	WeakSet
Values	Any type (objects + primitives)	Objects only
size property	Yes	No
Iteration	Yes (forEach, values, etc.)	No
Weak references	No	Yes
Garbage collection	Not automatic	Automatic

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript WeakSet</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>WeakSet vs Set</h4>
<p id="demo"></p>

<script>
// Set can hold any type
const set = new Set(["string", 42, true, {obj: true}]);

// WeakSet can only hold objects
const obj = {name: "John"};
const weakSet = new WeakSet([obj]);

let text = "<b>Set can hold:</b><br>";
text += "- Strings: ✓<br>";
text += "- Numbers: ✓<br>";
text += "- Booleans: ✓<br>";
text += "- Objects: ✓<br><br>";

text += "<b>WeakSet can only hold:</b><br>";
text += "- Objects: ✓ (only types allowed)<br><br>";

text += "<b>Set has:</b> size, forEach(), values(), keys()<br>";
text += "<b>WeakSet has:</b> add(), delete(), has() only<br><br>";

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>

```

Example 3

Result [View Example](#)

Weak References

WeakSet holds objects by **weak references**. When an object is no longer referenced elsewhere, it can be garbage collected:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript WeakSet</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Weak References</h4>
<p id="demo"></p>

<script>
const weakSet = new WeakSet();

let john = { name: "John" };
weakSet.add(john);

let text = "Object added to WeakSet<br><br>";

// Display object
text += "Object in WeakSet: " + (weakSet.has(john)) + "<br><br>";

// Remove all references to john
john = null;

text += "After setting john = null:<br>";
text += "The object can now be garbage collected<br>";
text += "It will be automatically removed from the WeakSet<br><br>";
text += "This is why WeakSet is useful for<br>";
text += "managing memory efficiently";

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

Example 4

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript WeakSet](#)